

Modeling and Design of Three-Phase Systems Using Complex Transfer Functions

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Abstract - *The use of complex vectors in the modeling, analysis and design of three phase AC systems is gaining wider use. Largely complex vectors have been used with coordinate transformations, signal flow diagrams, and in few cases with transfer functions, Root Loci and Bode plots. This paper formally and comprehensively extends complex vector representation to the standard set of control system tools including transfer functions, Laplace transforms, Nyquist and Bode plots, and stability criterion. The simple and insightful approach is developed and demonstrated in terms of the well known principles of control theory. A design example is given for a three phase AC voltage regulator with linear and non-linear unbalanced loading. Direct design and analysis of a complex controller with complex gains are performed based on the results of this paper. Simulation results further illustrate the complex regulator performance. Lastly experimental results validate the developments and confirm the simulation results.*

Introduction

The use of space vectors represented as complex numbers has gained widespread use in the analysis of three phase machines and systems. The compact representation of the three phase quantities provides a very elegant means for describing and analyzing such systems. The objective of this paper is to rigorously and comprehensively extend and demonstrate the use of complex vectors beyond that presently reported and into the familiar analysis and design domain established by control theory. The complex variable representation gives an elegant and compact means to deal with the analysis, design and representation of three phase systems, whether they are electric machines or pure power circuits and thus motivates the principle objective of this paper. Complex vector representation has been used to reduce the complexity of signal flow graphs and lend more insight into the physical nature of the system under study. This same simplicity has been demonstrated in the calculation and visualization of eigenvalues and their respective root loci plots. Carrying the notions further, this paper, rigorously reviews and develops the complex transfer functions and extends their analysis into the realm of conventional control theory that is so pervasive in the analysis of closed loop systems. Again the benefits of added simplicity and additional insight are gained as the foundations of control theory are naturally extended to complex signal vectors while maintaining the familiar concepts of real variable analysis.

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To the objective of this study, simple linear transformations are first defined in review of the coordinate transformations from abc coordinates into the synchronously rotating dq coordinates. This provides a solid theoretical base for subsequent developments, from which the formal definition of complex transfer functions are derived. Throughout the formulation, the reader is kept mindful of the real three phase variable representations with illustrations of how the complex form can be easily unfolded into the familiar real variable format. With complex transfer functions formally defined, the development proceeds to application of standard control theory to the complex formulation. Laplace transforms of the complex systems, signals and their derivatives are formulated where the notion of a local frequency domain relative to the local coordinate system is conveniently defined. Making use of conformal mapping and the Nyquist stability criterion, which places no restriction on the real or imaginary nature of the system, the Nyquist stability criterion is rigorously generalized for the complex transfer functions. Stemming from this development, complex forms of Nyquist, and Bode plots are illustrated. The complex nature appears to result in asymmetrical frequency properties as compared to the real variable counterparts. The apparent asymmetrical frequency nature is inherent to the local frequency domain definition and lends critical insight into the dynamic behavior and design of three phase controllers. The result of these developments is a powerful extension of classical control theory for application to three phase systems that elegantly parallels the classical analysis and design of real variable control systems. To demonstrate the power and utility of these tools, the step by step design of an unbalanced three phase voltage regulation system is given. The concept and utility of a complex controller with complex gains becomes evident as well as its simple real variable hardware/software implementation. Simulation and experimental results from an actual three phase system validate the results and demonstrate excellent dynamic performance.

Three-Phase Models

Consider first a three phase inductor shown in the Fig. 1. The equation that describes the inductor in the abc coordinate system, shown in Fig. 1, is:

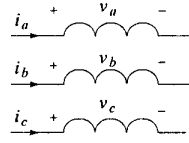


Fig. 1. Three-phase inductor

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = L \frac{\partial}{\partial t} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}. \quad (1)$$

Applying a linear transformation,

$$S = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad (2)$$

changes the abc nonorthogonal coordinate system into the orthogonal $\alpha\beta 0$ coordinate system, shown in Fig. 2. The inductor equation in the $\alpha\beta 0$ coordinates is:

$$\begin{bmatrix} v_\alpha \\ v_\beta \\ v_0 \end{bmatrix} = L \frac{\partial}{\partial t} \begin{bmatrix} i_\alpha \\ i_\beta \\ i_0 \end{bmatrix}. \quad (3)$$

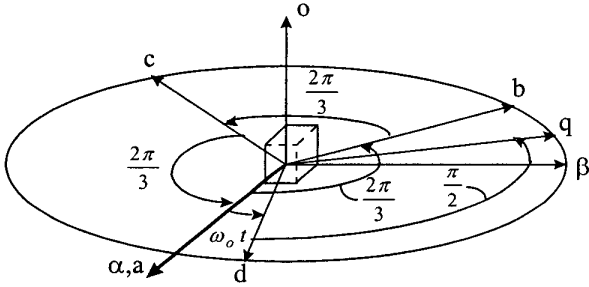


Fig. 2. Reference coordinate systems

If the currents i_a , i_b , and i_c , are periodical with frequency ω_b , then the coordinate system can be changed from rotational (synchronous) dq0 to stationary $\alpha\beta 0$, also shown in Fig. 2, by applying the following transformation.

$$T_{R2S} = \begin{bmatrix} \cos(\omega_0 t) & -\sin(\omega_0 t) & 0 \\ \sin(\omega_0 t) & \cos(\omega_0 t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

Then equation (3) becomes:

$$T_{R2S} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = L \frac{\partial}{\partial t} \left(T_{R2S} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \right) \quad (5)$$

After some rearrangements and calculations:

$$\begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = L \left(\begin{bmatrix} 0 & -\omega_0 & 0 \\ \omega_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \frac{\partial}{\partial t} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \right) \quad (6)$$

The same equation in the state space form is:

$$\frac{\partial}{\partial t} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \frac{1}{L} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} + \begin{bmatrix} 0 & \omega_0 & 0 \\ -\omega_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \quad (7)$$

The equation (7) equivalent block diagram is shown in Fig. 3.

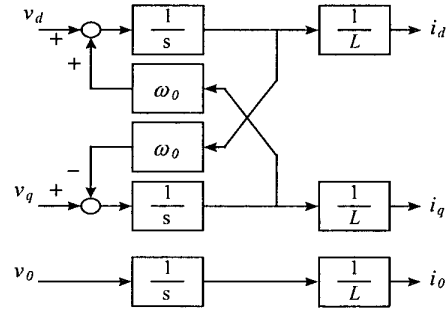


Fig. 3. Block diagram of a three-phase inductor in rotational, dq0, coordinates

Applying a linear transformation,

$$W = \begin{bmatrix} 1 & j & 0 \\ 1 & -j & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (8)$$

to equation (5), yields:

$$W T_{R2S} W^{-1} W \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = L \frac{\partial}{\partial t} \left(W T_{R2S} W^{-1} W \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \right), \quad (9)$$

which after some algebra leads to (10). The first and second rows of equation (10) are two complex conjugate equations. Due to the fact that the matrices are diagonalized, the second row can be omitted which gives (11). Equation (11) can be also put in the state space form (12). Equation (12) is a set of two independent equations. The first one is a complex equation describing the relationships in dq plane and the second one is zero sequence equation. Fig. 4. shows the block diagram of the equation (12).

$$\begin{bmatrix} e^{j\omega_0 t} & 0 & 0 \\ 0 & e^{-j\omega_0 t} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_d + jv_q \\ v_d - jv_q \\ v_0 \end{bmatrix} = L \frac{\partial}{\partial t} \begin{bmatrix} e^{j\omega_0 t} & 0 & 0 \\ 0 & e^{-j\omega_0 t} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_d + ji_q \\ i_d - ji_q \\ i_0 \end{bmatrix}. \quad (10)$$

$$\begin{bmatrix} e^{j\omega_0 t} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_d + jv_q \\ v_0 \end{bmatrix} = L \frac{\partial}{\partial t} \begin{bmatrix} e^{j\omega_0 t} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_d + ji_q \\ i_0 \end{bmatrix}. \quad (11)$$

$$\frac{\partial}{\partial t} \begin{bmatrix} i_d + ji_q \\ i_0 \end{bmatrix} = \frac{1}{L} \begin{bmatrix} v_d + jv_q \\ v_0 \end{bmatrix} - \begin{bmatrix} j\omega_0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} i_d + ji_q \\ i_0 \end{bmatrix}. \quad (12)$$

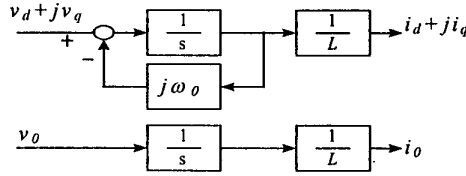


Fig. 4. Block diagram of a three-phase inductor in rotational, dq0, coordinates in complex notation

It is important to notice that signals of the dq part of the block diagram are complex. Fig. 4. can be unfolded into the Fig. 3. These two block diagrams are exactly the same. The equivalent three phase capacitor block diagram is shown in Fig. 5.

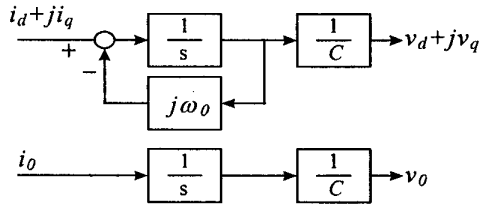


Fig. 5. Block diagram of a three-phase capacitor in rotational, dq0, coordinates in complex notation

Fig. 4. and Fig. 5. give more than just complex notation, the equivalent transfer functions (13) and (14) can be found. Equations (13) and (14) represent a new class of transfer functions - polynomial transfer functions of a complex variable with complex coefficients. In further text, these transfer functions will be referred as "complex transfer functions", while polynomial transfer functions of a complex variable with real coefficients will be referred to as "real transfer functions".

$$G_L = \frac{i_d + ji_q}{v_d + jv_q} = \frac{i_{dq}}{v_{dq}} = \frac{1}{(s + j\omega_0)L}, \quad (13)$$

$$G_C = \frac{v_d + jv_q}{i_d + ji_q} = \frac{v_{dq}}{i_{dq}} = \frac{1}{(s + j\omega_0)C}. \quad (14)$$

Laplace Transform in Rotational Coordinates

Before the complex transfer functions are further examined, it is necessary to exam the Laplace transform in the rotational coordinates.

$$F_{dq}(s_{dq}) = L(f_{dq}(t)) = L(f_{\alpha\beta}(t)e^{-j\omega_0 t}) = F_{\alpha\beta}(s_{dq} + j\omega_0) \quad (15)$$

Laplace transform expression, (15), gives the relationship between the Laplace transform in the rotational and stationary coordinates. The Laplace transform in the rotational coordinates can be calculated by replacing every occurrence of the Laplace operator, s , in the stationary transform by $s_{dq} + j\omega_0$, where ω_0 is the rotational frame angular speed. Please note that the equation (15) imaginary part of the operator s_{dq} is frequency with respect to the rotational coordinates. The purpose of the subscript dq is to emphasize that. By analogy the Laplace transform in the stationary coordinates is calculated in terms of the Laplace transform in rotational coordinates, as shown in equation (16).

$$F_{\alpha\beta}(s_{\alpha\beta}) = L(f_{\alpha\beta}(t)) = L(f_{dq}(t)e^{j\omega_0 t}) = F_{dq}(s_{\alpha\beta} - j\omega_0) \quad (16)$$

The subscript $\alpha\beta$ of the complex frequency $s_{\alpha\beta}$ should emphasize that its imaginary part is frequency with respect to the stationary frame.

It is interesting to analyze the relationships between the Laplace transforms of the first derivative of a function in the stationary and rotational coordinates, as shown in (17).

$$\begin{aligned} s_{dq} F_{dq}(s_{dq}) - f_{dq}(0) &= L\left(\frac{\partial}{\partial t} f_{dq}(t)\right) = \\ L\left(\frac{\partial}{\partial t} (f_{\alpha\beta}(t)e^{-j\omega_0 t})\right) &= \\ L\left(\left(-j\omega_0 f_{\alpha\beta}(t) + \frac{\partial}{\partial t} (f_{\alpha\beta}(t))\right)e^{-j\omega_0 t}\right) &= \\ s_{dq} F_{\alpha\beta}(s_{dq} + j\omega_0) - f_{\alpha\beta}(0) \end{aligned} \quad (17)$$

Laplace transform of the derivative of a function in rotational coordinates can be calculated by replacing all occurrences of s in the Laplace transform of the function in stationary frame by $s_{dq} + j\omega_0$, multiplication by s_{dq} , and subtraction of the initial condition. Equations (15) and (17) give us a way to effortlessly write the system equations in rotational coordinates by looking into the single phase complex equivalent of a real three-phase system.

Complex Transfer Function

The system stability is the most important issue in every control system design. The stability analyses for the complex transfer functions is not much different from that of real transfer functions. The basis of the stability analyses is the theory of conformal mapping [1].

$$TF = \frac{G(s)}{1 + T(s)} \quad (18)$$

Equation (18) is the most general form of a transfer function.

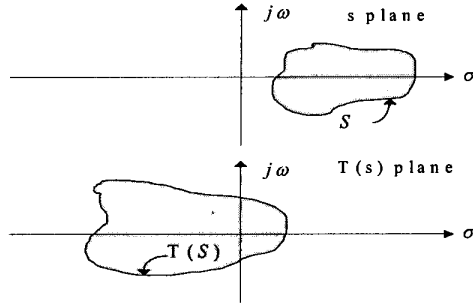


Fig. 6. Conformal mapping of regions in s plane onto $GH(s)$ plane

If the function $T(s)$ is analytical (it has defined first derivative for each point regardless of the direction of the approach) on the contour S , the value of $T(s)$ can be calculated for the each point on the contour S and plotted in the $T(s)$ plane. The result is a contour, $T(S)$, in the $T(s)$ plane. If the $T(s)$ is analytical for every point inside the contour S , then each point inside the S has corresponding point inside the contour $T(S)$ in the $T(s)$ plane. If it is not the contour S is modified such that the poles are excluded, e.g. placed outside of the contour S . The theory is valid for both real and complex transfer functions. If the contour S is chosen such that it covers the entire right-half-plane, RHP, the system stability can be analyzed. The characteristic equation is $1+T(s)$, so the $T(S)$ contour must not contain point -1 . Otherwise there is a point in RHP for which the characteristic equation is equal zero and the system is unstable. Using the principle of this argument, Nyquist has derived the famous Nyquist stability criteria [2]: The number of closed loop poles in the RHP is equal to the sum of the number of open loop poles in the RHP and the number of clockwise encirclements of the -1 point on the $GH(s)$ plane, (19).

$$N_{CL} = N_{OL} + N_{CW_ENC_OF_ -1} \quad (19)$$

The proof for the Nyquist criteria also does not assume that the transfer function is real. The only place where this is assumed is during the plotting of the Nyquist plot. Only the positive half of the Nyquist plot is plotted due to the symmetry that is a consequence of the real transfer functions. This means that all tools of classic control theory can be applied to the complex transfer functions.

Laplace transform expression (15) gives us a clue how to interpret equation (13) and (14). It can be seen in (15) that transfer function $F_{dq}(s_{dq})$ is a real transfer function of operator $(s_{dq} + j\omega_b)$. Further, it means that the Nyquist plot of transfer function in the stationary and rotational frames are exactly the same and symmetrical around the real axis. However, the frequency calibration of the Nyquist plot in rotational coordinates is changed. If two symmetrical points on the Nyquist plot are chosen, they do not correspond any more to the positive and negative value of the same frequency. There is a frequency skew of ω_b . An important conclusion is that the stability does not depend on the reference system. Stability margins will be the same in all reference systems, the only value that can change is the crossover frequency, which is affected by the frequency shift of $-j\omega_b$. To illustrate this a three-phase inductor Nyquist plot is shown in Fig. 7. Equation (13) can be rearranged as

$$G_L = \frac{1}{(\sigma + j(\omega + \omega_o))L} = \frac{1}{s_{dq}L}. \quad (20)$$

One can see that a part of the Nyquist plot what would correspond to the frequencies $\omega = (0, \infty]$ and $\omega = (-\infty, 0]$, in dq frame correspond to $\omega = (-\omega_b, \infty]$ and $\omega = [-\infty, -\omega_b)$, respectively. It should be noted that the frequency ω in equation (20) is the frequency seen from the dq frame.

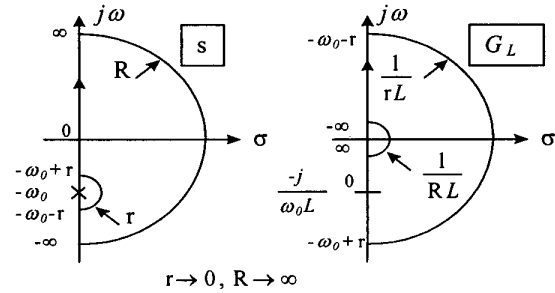


Fig. 7. The Nyquist plot of the transfer function of a three phase inductor in dq frame given by equation (20)

The Bode plot of equation (20) is shown in Fig. 8. The first impression is that the plot is not symmetrical around $-\omega_b$. There is no pole at ω_b , the flat part between $-\omega_b$ and ω_b is a consequence of the logarithmic scale. If the plot is plotted using linear frequency scale, it would be seen that it is symmetrical around $-\omega_b$.

Design Procedure

The design procedure will be demonstrated using an example. The control law for an inverter with three phase LC filter and neutral wire will be designed. This topic has gained recent attention [3] with the expanding uninterruptible power supplies market. Fig. 9. shows a three phase LC filter and its model in the dq frame. According to equation (12) there are two independent parts. The first one is the complex domain dq model and the other is the zero sequence model. All

variables in the dq model are complex and defined according to $x_{dq} = x_d + jx_q$. All variables in zero

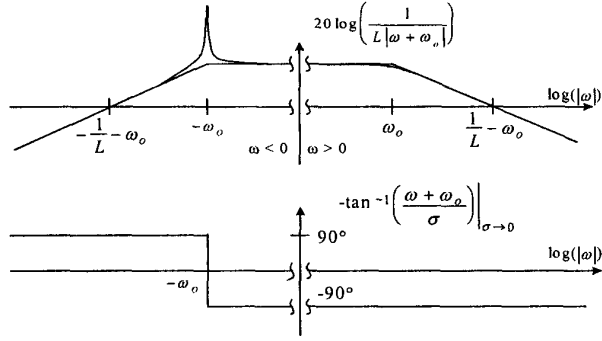


Fig. 8. The Bode plot of the transfer function of a three phase inductor in dq frame given by equation (20)

sequence model are real. Due to the symmetry the gains in the zero sequence regulator will be real parts of the gains in dq regulator, which further simplifies the design procedure.

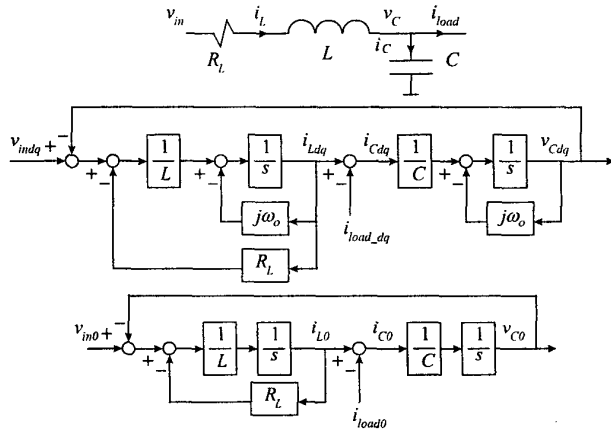


Fig. 9. Model of a three phase LC filter in rotational dq frame.

The first step is the current loop design. Proportional regulator with feed forward and state feedback decoupling is chosen as the current regulator, as shown in Fig. 10.

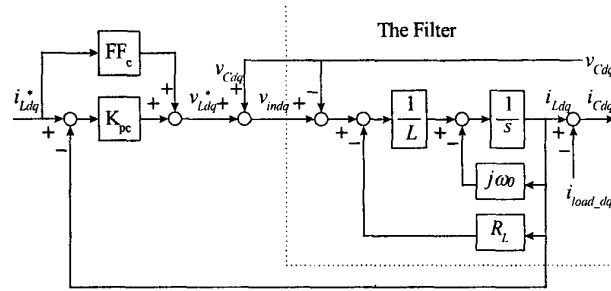


Fig. 10. The current loop of an inverter with an output LC filter

State feedback decoupling of the output capacitor voltage eases the current loop design as control voltage effectively only drives the inductor. It is obvious to see in Fig. 10. that

the decoupling cancels internal filter feedback. The resulting characteristic equation is:

$$1 + K_{pc} \frac{1}{sL + j\omega_0 L + R_L} = 0 \quad (21)$$

Direct design allows calculation of the controller complex gains given the desired eigenvalues. Replacing s with $s_{desired}$ and solving the characteristic equation for K_{pc} we get:

$$K_{pc} = -s_{desired} L - j\omega_0 L - R_L \quad (22)$$

which gives the exact value for the proportional gain. We can place the current loop pole at any place, equation (22) still holds. If one complex pole is placed, unfolding the system as shown in Fig. 3 will reveal the complex conjugate pole, as legacy of the second row of equation (10). If a pole on the real axes is chosen, the system will have double real pole after unfolding. The feedforward equation is:

$$FF_c \frac{1}{sL + j\omega_0 L + R_L} = 1, \quad s = 0. \quad (23)$$

By replacing s with 0, and solving for FF_c we get:

$$FF_c = j\omega_0 L + R_L. \quad (24)$$

Fig. 11. shows the implementation of the complex gains. The block diagrams in Fig. 11. are derived from equations (25).

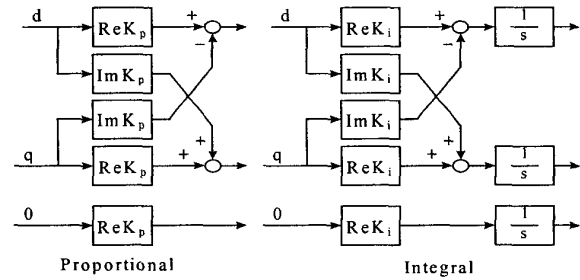


Fig. 11. The complex gains implementation

$$\begin{aligned} (d + jq)(\text{Re } K_p + j \text{Im } K_p) &= \\ (d \text{Re } K_p - q \text{Im } K_p) + j(d \text{Im } K_p + q \text{Re } K_p) \\ \int (d + jq)(\text{Re } K_i + j \text{Im } K_i) dt &= \\ \int (d \text{Re } K_i - q \text{Im } K_i) dt + j \int (d \text{Im } K_i + q \text{Re } K_i) dt \end{aligned} \quad (25)$$

Derivative gain block diagram is the same as integral, the only difference is that "1/s" is replaced by "s". In the case of fixed frequency output all gains are constants, but if the frequency varies some are not. Under the assumption that the output frequency does not change within a switching cycle, all gains are still valid. Implementation of the controller in dq coordinates requires a microprocessor so it is easy to recalculate variable gains every switching cycle. This will

ensure that poles of the system do not move with change of the output frequency.

The complex block diagram of the voltage loop is shown in Fig. 12.

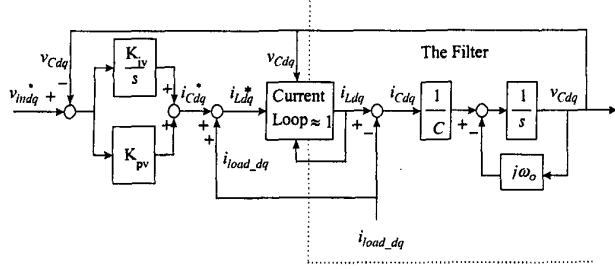


Fig. 12. The voltage loop of an inverter with an output LC filter

Following the principle of the pole separation, the current loop bandwidth should be designed such that it is at least 6 to 7 times faster than the voltage loop. In that case the current loop transfer function may be approximated to be equal unity. As with the current loop, it is easier to design the gains if the load current is decoupled from the output capacitor. It is obvious to see in Fig. 12. that the decoupling cancels the internal filter feedback. The characteristic equation is:

$$1 + \left(K_{pv} + \frac{K_{iv}}{s} \right) \frac{1}{sC + j\omega_o C} = 0 \quad (26)$$

By replacing s the first time by $s_{1desired}$ and the second time by $s_{2desired}$ in equation (26) two equations are created. They can be solved for the gains K_{pv} and K_{iv} .

$$\begin{aligned} K_{pv} &= -(s_{1desired} + s_{2desired})C - j\omega_o C \\ K_{iv} &= s_{1desired}s_{2desired}C \end{aligned} \quad (27)$$

The two voltage loop poles can be placed anywhere. Everything said for the current loop poles also applies to the voltage loop poles. The Fig. 13. shows the complete block diagram of the control loops of an inverter with output LC filter. All gains have the structure shown in Fig. 11, and all variables on the left of the coordinate transformations are two dimensional vectors, as in equation (12). The first element is complex dq variable and the second one is real zero sequence variable. Variables on the right of the coordinate transformations are three dimensional real vector in abc coordinates. Fig. 11. assumes ideal voltage modulator and inverter with a transfer function of unity.

Simulation Results

An inverter with three-phase LC filter was simulated. The parameters used in the simulation are: filter inductance $L = 40 \mu\text{H}$, inductor resistance $R_L = 1 \text{ m}\Omega$, filter capacitor $C = 20 \mu\text{H}$, output frequency $f_o = 400 \text{ Hz}$, current loop pole at 10 kHz , double voltage loop pole at 1800 Hz , switching frequency $f_{sw} = 62.5 \text{ kHz}$, e.g delay time in the control loop is $T_D = 16 \text{ ms}$, input dc voltage $V_{DC} = 350 \text{ V}$, and output voltage

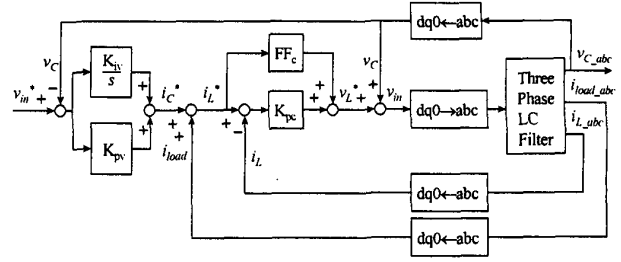


Fig. 13. The control loops of an inverter with an output LC filter $V_o = 165 \text{ V}$ peak. Voltage drop across switches, dead time distortions and parasitic effect are neglected in the simulations. Fig. 14. shows the load current, output voltage in abc coordinates, and output voltage in dq0 coordinates for a linear unbalanced load. It can be seen that control loop handles unbalanced linear load perfectly.

Fig. 15. shows the load current, output voltage in abc coordinates, and output voltage in dq0 coordinates for a nonlinear unbalanced load. The nonlinear load consists of three diode bridge rectifiers connected between phase and neutral wire draining the peak current of nearly 20 A , a three-phase diode bridge rectifier draining the peak current of nearly 15 A , and a linear load draining 7 A peak.

Fig. 16 illustrates the simulation results for a line started induction motor (50 Hp , 230 Volt , 4 pole 60 Hz Induction machine). Over current protection of the inverter required line starting at a reduced voltage. The simulation results indicate that the line voltages remain undistorted during this transient load condition.

Experimental Results

Fig. 17 illustrates the experimental results for the line started motor transient. The experimental results agree very well with the simulation results of Fig. 16. To further investigate the performance of the controller, a non-linear three phase rectifier load is added to the output. Fig. 18 illustrates the output line voltages and line current at rated voltage. The motor is loaded to 120 amps peak and the diode bridge is loaded with $2700 \mu\text{F}$ and 6.75 Ohms . The waveforms illustrate low distortion with a THD of 6.4% .

Conclusion

This paper has extended the classical control system analyses and design techniques to include complex signal representations common to three phase analysis. Specifically, transfer functions, Laplace transforms, Nyquist plots and stability criterion are derived and illustrated in term of the complex signals. This analysis results in an elegant and simple representation of three phase control systems. For illustration, a three phase AC voltage regulator is designed and implemented in simulation and experimentally.

Acknowledgments

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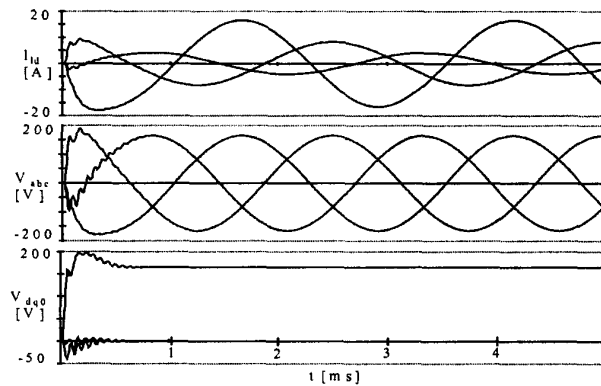


Fig.14. The load current, output voltage in abc, and output voltage in dq0 coordinates for an unbalanced linear load

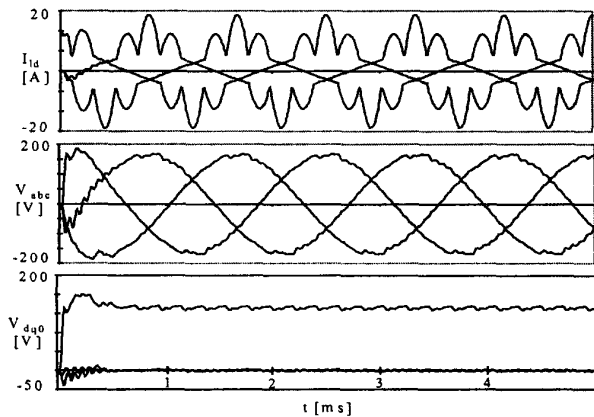


Fig.15. The load current, output voltage in abc, and output voltage in dq0 coordinates for an unbalanced nonlinear load

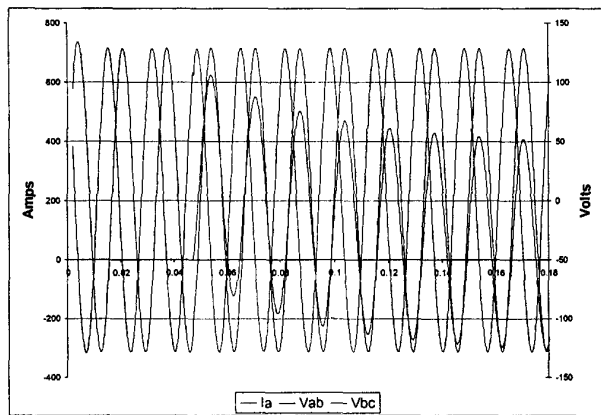


Fig.16. Simulation of induction motor line start transient

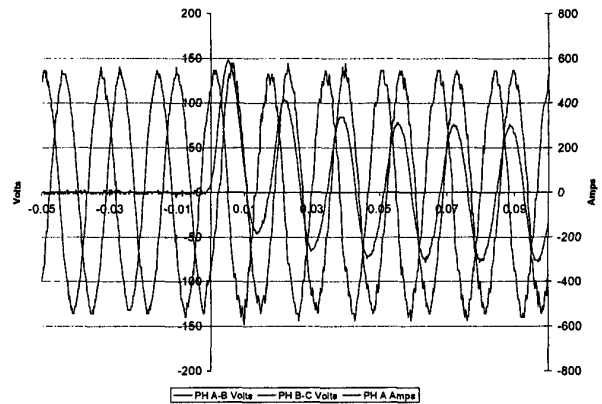


Fig.17. Experimental result of induction motor line start transient

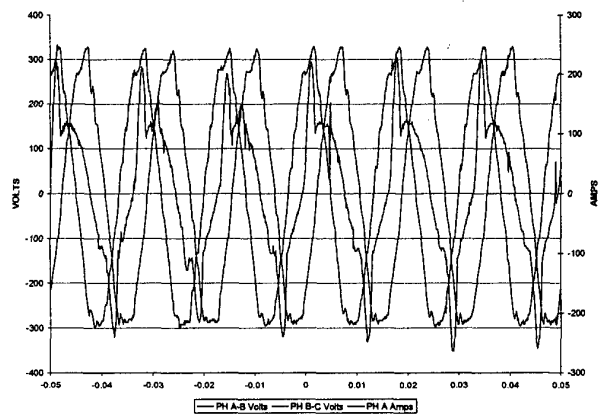


Fig.18. Experimental results of IM and non linear load

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